

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	10 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form
FR-2	User Login	Login with email and password, JWT token generated
FR-3	Product Catalog	View all products, filter by category, search by name
FR-4	Cart Management	Add product to cart, view cart, remove from cart
FR-5	Order Management	Place order from cart, view order history
FR-6	Admin Dashboard	Add product, edit product, delete product, view all orders

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The app has a clean and simple UI built with Tailwind CSS that any user can navigate without confusion
NFR-2	Security	Passwords are hashed using bcryptjs and all protected routes use JWT token authentication
NFR-3	Reliability	The app handles errors gracefully and shows appropriate messages when something goes wrong
NFR-4	Performance	API calls are made using Axios and MongoDB queries are fast for small to medium data sets
NFR-5	Availability	The app runs locally and can be deployed to Vercel and Render for 24/7 availability
NFR-6	Scalability	Frontend and backend are decoupled so they can be scaled independently as the user base grows